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Preterm birth aetiology 2004–2008. Maternal factors associated with three phenotypes: spontaneous preterm labour, preterm pre-labour rupture of membranes and medically indicated preterm birth

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Objectives: **To (1) investigate the current distribution of PTB phenotypes; (2) identify factors associated with spontaneous preterm labour (SPTL), PPROM, and indicated PTB; (3) investigate the relationship of gestational age (ga) with each PTB phenotype.** ***Methods:*** **Retrospective review of all live, singleton births 23^{+0} to 36^{+6} weeks ga at an obstetric referral centre 2004–2008.** ***Results:*** **A total of 4,522 PTBs were included (SPTL 31.7%, PPROM 27.4%, indicated 40.8%). PTB phenotype distribution differed between ga groups (<27 weeks: SPTL 45%, PPROM 32%, indicated 23%; 27–33 weeks: SPTL 30%, PPROM 32%, indicated 39%; 34–36 weeks: SPTL 32%, PPROM 24%, indicated 44%, $p < 0.001$). Between 34–36 weeks', demographic factors were significantly different between PTB phenotypes (age ≥35: SPTL 13.8%, PPROM 15.4%, indicated 21.6%; Caucasian ethnicity: SPTL 61.6%, PPROM 69.0%, indicated 70.2%; Assisted Reproductive Technology (ART): SPTL 2.8%, PPROM 1.9%, indicated 9.3%; all $p < 0.001$). Between 27–33 weeks' PTB phenotype was associated with smoking (SPTL 24.9%, PPROM 29.3%, indicated 20.2%; $p = 0.002$) and ART (SPTL 2.3%, PPROM 1.6%, indicated 5.0%; $p = 0.002$). Demographic factors were not associated with PTB phenotype at 23–26 weeks.** ***Conclusions:*** **The increase in PTB rates may be explained by medical indications at late preterm gestations, primarily in older, Caucasian women requiring fertility assistance. Interventions to reduce the rate of PTB need to be targeted to this high-risk population.**

Keywords: etiology, obstetric labour, pregnancy, pregnancy complications, premature, premature birth

Introduction

Preterm birth (PTB) is the leading cause of perinatal morbidity and mortality and represents a significant burden on the global health care system [1,2]. Worldwide, the rate of PTB is rising and currently represents 9.6% of all births. Neonatal management has made significant progress in recent years and this has resulted in substantial improvement in the outcomes of preterm babies, in particular babies born at the margins of viability. In vast contrast, obstetric and public health interventions aimed at preventing PTB have so far been relatively ineffective [3].

In Australia, as in other developed countries, preterm infants born at late gestational ages (34–36 weeks) account for most of the increases in PTB rates [1,4]. Although not consuming the same level of resources as early PTBs, evidence is emerging that these infants have a substantially raised risk of morbidity compared with term infants [5–7]. The short- and long-term sequelae of late PTB, including increased neonatal intensive care admissions and adverse neurodevelopmental outcomes, contribute to the growing overall cost of PTB.

PTB has long been known to be a complex clinical syndrome with multiple aetiologies and presentations [8–10]. In spite of this, many scientific studies continue to regard PTB as a single end-point: birth before 37 weeks of gestation, hindering the application of many findings [11–13]. The failure to account for PTB heterogeneity has limited our understanding of its causes and may have contributed to ineffective interventions to prevent PTB.

PTB is traditionally characterised into three clinical phenotypes: spontaneous labour with intact membranes (SPTL), preterm pre-labour rupture of membranes (PPROM) and PTB due to maternal or fetal complications without spontaneous labour. It is estimated that about 40–45% of PTB are due to SPTL, 25–30% due to PPROM and 30–35% are indicated PTBs [8,14]. The most recent Australian study of which we are aware, conducted in 1988, found that indicated PTB comprised 26% of all live, very low birthweight infants [15].

A number of risk factors have been linked to PTB in general, including ethnicity [8], social disadvantage [8,11], maternal age, infertility [16], multiple births, previous PTB [17], stress [18], maternal smoking [13] and genetic factors [12]; however, results from epidemiological studies have been inconsistent. This is likely due to poor phenotype characterisation as well as variations in the populations studied.

In the context of rising PTB rates [1,3], it is timely to re-investigate the distribution of PTB using clearly defined phenotypes.

The aims of the present study were: 1) to investigate the current distribution of PTB; 2) to identify factors associated with three preterm birth phenotypes – SPTL, PPROM and medically indicated PTB; and 3) to investigate the relationship of gestational age with each PTB phenotype.

Methods

The study was a retrospective cohort study of preterm births at an Australian hospital during the 5 year period 2004–2008. All live, singleton preterm births between 23^{+0} and 36^{+6} weeks of gestation were included. Exclusion criteria included multiple pregnancies, stillbirths and terminations of pregnancy due to lethal

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fetal anomalies. The study hospital is the only tertiary-referral obstetric centre in the state of Western Australia and manages the majority of complicated pregnancies from a large geographic area. Women who present to metropolitan and rural centres in the state with anticipated early PTB are routinely transported to this hospital before the baby is born unless birth is imminent. The rationale for this policy is based on extensive evidence indicating improved morbidity and reduced mortality in PTB infants born in a tertiary referral hospital compared with outborn infants [19]. Consequently, most early preterm births as well as a large proportion of the state's late preterm births occur in this institution.

An extract of de-identified data for all preterm births was obtained from the centre's electronic perinatal database, which contains comprehensive obstetric information on all births greater than 20 weeks' gestation. Data from the state-wide birth notification system were used for comparison and validation of the data.

Gestational age was determined by the date of the last known menstrual period with confirmation from first trimester ultrasound biometry. If dates were not consistent with the ultrasound estimation or not known, the first trimester ultrasound biometry alone was used to determine the estimated date of delivery. If the first trimester ultrasound was not available the gestational age was determined by the date of the last known menstrual period or an ultrasound performed in a later trimester.

Spontaneous preterm labour (SPTL) was defined as spontaneous onset of labour with intact membranes and delivery before 37 completed weeks of gestation. PPROM was defined as the spontaneous rupture of the fetal membranes before the onset of labour earlier than 37 weeks of gestation. Indicated PTB was defined as any planned preterm birth in which labour was induced or the delivery was by elective caesarean section for a medical indication without spontaneous onset of labour or PPROM. All obstetric or fetal complications for which early delivery was considered necessary were included.

We defined three gestational age phenotypes a priori, extreme PTB (23^{+0} to 26^{+6} weeks'), very PTB (27^{+0} to 33^{+6} weeks') and late PTB (34^{+0} to 36^{+6} weeks'). Continuous variables were summarised as mean (95% confidence interval) or median (range) as appropriate and compared using ANOVA. Categorical data were compared using either Fisher's Exact Test or the χ^2-test as appropriate. The accepted level of significance was 0.01 for all analyses in order to account for multiple comparisons.

This study was given approval by the ethics committee of the Women's and Newborn's Health Service. De-identified data only were utilised in this study.

Results

A total of 27,951 births were registered at the study hospital between 2004 and 2008. Preterm birth was identified in 5469 (20.9%) births. After exclusions 4522 live births from a singleton pregnancy were included in the study (Figure 1). SPTL, PPROM and indicated PTB occurred in 31.7%, 27.4%, and 40.8% of PTBs respectively.

PTB numbers increased over the study time period in conjunction with an overall increase in births. The number of late PTBs increased from 428 in 2004 to 570 in 2008. The median gestational

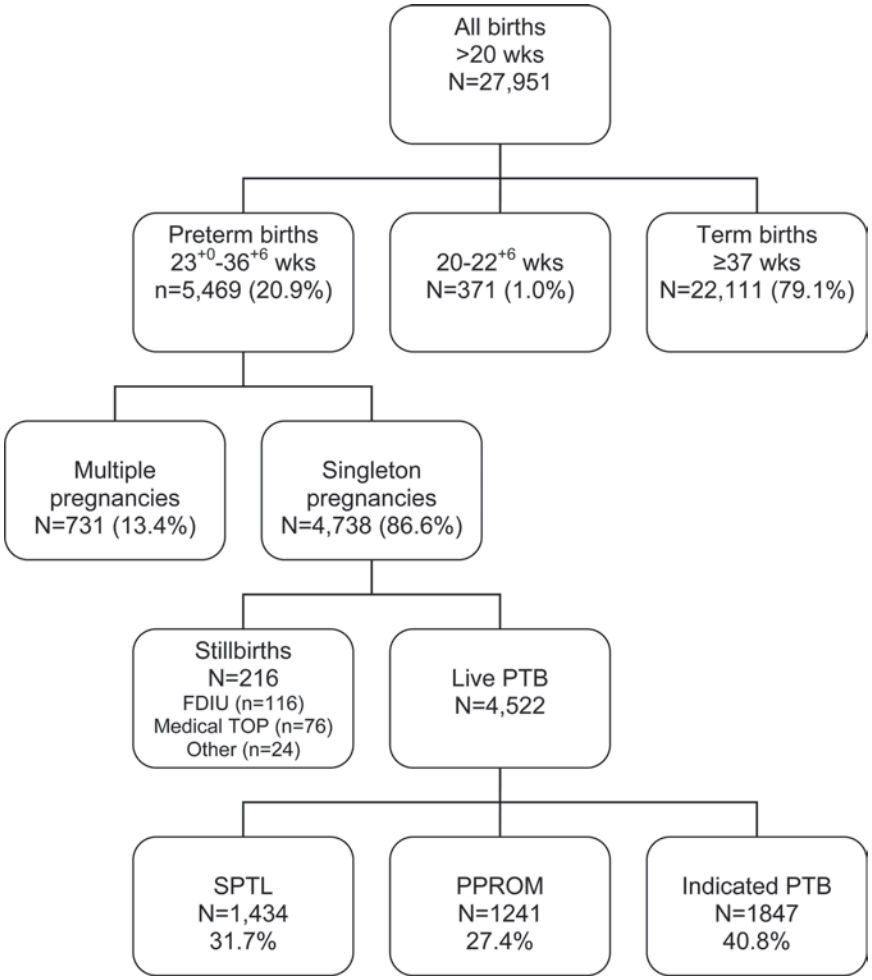

Figure 1. Preterm births 2004–2008.

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age of the PTBs was 34 weeks. The distribution of gestational age groups was 6%, 36% and 58% for extreme PTB, very PTB and late PTB respectively. These proportions did not change over the five year period ($p = 0.375$). The numbers of births represented 92%, 85% and 39% of extreme PTB, very PTB and late PTB respectively of live, singleton PTB notifications for the region in the same time period.

The distribution of PTB phenotype changed significantly over the five year period (Figure 2, $p = 0.001$). Indicated PTB increased from 39.5% of all PTBs in 2004 to 44.0% in 2008. In contrast the proportion of PTB due to PPROM decreased from 30.8% to 22.8%. Overall, PTB phenotype varied significantly between gestational age groups (Figure 3, $p < 0.001$). SPTL occurred in 44.7% of births at extremely early gestational age. In contrast, medically indicated PTB was the most prevalent phenotype at late gestational ages (44.1%).

The mean maternal age was 28.5 (6.5) years. There was a significant variety in the maternal demographic, pregnancy and delivery characteristics between gestational ages and PTB phenotypes. The greatest differences in demographic and pregnancy characteristics between PTB phenotypes occurred at late preterm gestational ages.

Characteristics of late PTBs

Between 34 and 36 weeks, women who had a medically indicated PTB were significantly older than women whose PTB was due to SPTL or PPROM (Table I). Women who had indicated PTBs were also more likely to be Caucasian and to have required assisted reproductive technology (ART). In contrast, women who had PPROM were more likely to be nulliparous than women with spontaneous PTL or medically indicated PTB.

As expected there were significant differences in the major complications of pregnancy between PTB phenotypes at late gestational ages (Table I). There were significantly greater proportions of women with pre-eclampsia and gestational diabetes in the indicated PTB phenotype compared with SPTL or PPROM. In

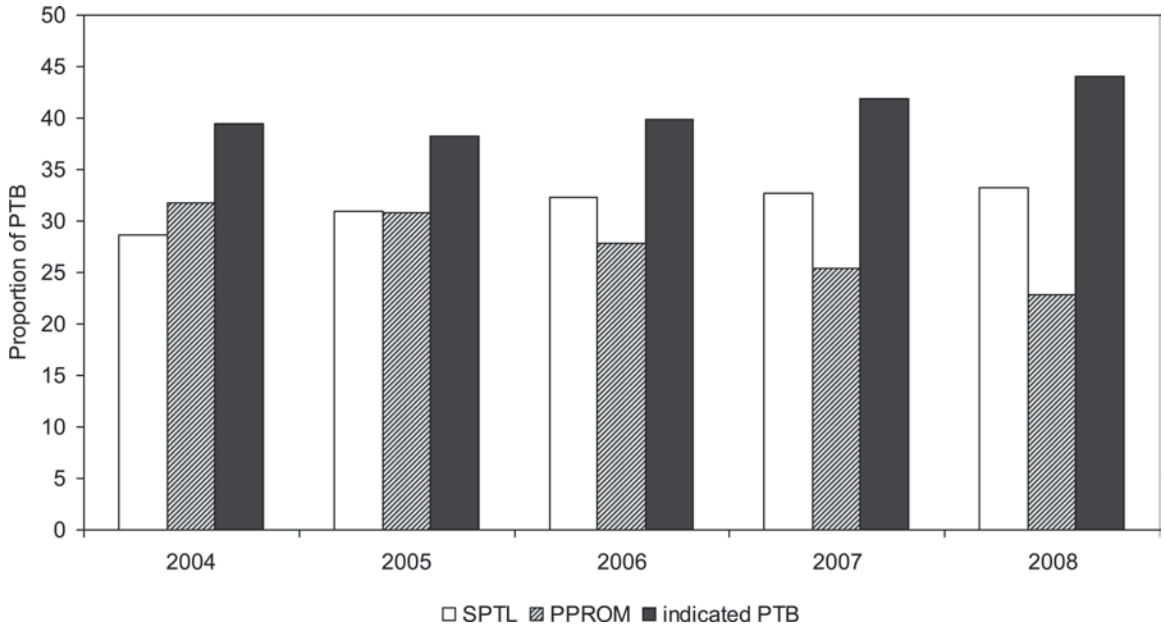

Figure 2. Distribution of PTB phenotypes over the 5-year period 2004–2008.

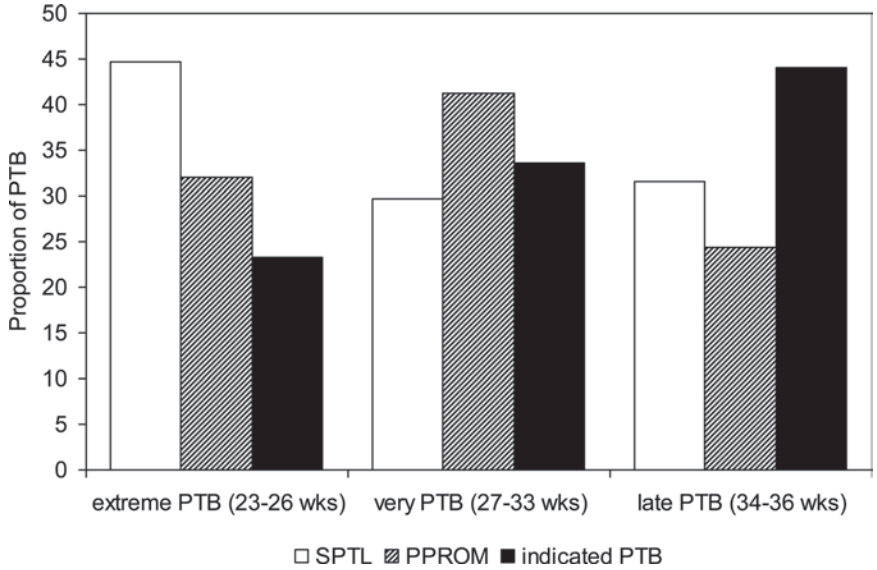

Figure 3. PTB phenotype within gestational age groups.

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addition the mean birthweight of SPTL infants was significantly greater that that of PPROM or indicated PTB infants suggesting an increased likelihood of fetal growth restriction in the latter two phenotypes. There were no differences in the proportions of women with placenta praevia or abruption between phenotypes; however a greater proportion of women with SPTL had a non-specified antepartum haemorrhage (APH) compared with other phenotypes (Table I).

Characteristics of very PTBs

Similar to late PTB, between 27 and 33 weeks women who had indicated PTBs were significantly more likely to have received ART than other PTB phenotypes. There were no differences in maternal age, ethnicity or parity, however, between PTB phenotypes (Table II). In contrast, women who had PPROM were more likely to smoke tobacco than women who had either spontaneous PTL or medically indicated PTB.

Over 50% of indicated PTBs in this gestational age group were associated with pre-eclampsia compared with small proportions in the other two phenotypes. Placental abruption was more prevalent in the indicated PTB phenotype. In contrast, other non-specified APH occurred more frequently in women with SPTL. Between 27 and 33 weeks, mean birthweight was significantly lower in the indicated PTB phenotype compared with SPTL or PPROM (Table II).

Characteristics of extreme PTBs

The demographic characteristics of women who delivered between 23 and 26 weeks were similar for all three PTB phenotypes. The prevalence of pre-eclampsia was 46.9% in the indicated PTB phenotype compared with minimal prevalence in the other two phenotypes. In contrast 30.9% of the SPTL phenotype had a non-specified APH compared with 9% of indicated PTBs. Similar to very PTB, the mean birthweight was significantly lower in the indicated PTB phenotype of extreme PTB.

Discussion

Preterm birth is a significant and increasing burden on our health-care system. To date, interventions to decrease the rate of PTB have not been successful [20]. Indeed, PTB rates are increasing world-wide. Our failure to fully elucidate the mechanisms of PTB contributes to this problem [3]. Study designs of many clinical and laboratory studies have not taken into account the heterogeneous nature of PTB, thus limiting our understanding of PTB in general.

The present study found substantial differences in demographic and pregnancy characteristics between clinical phenotypes of PTB. These differences were also apparent in PTBs at different gestational ages. At late gestational age, indicated PTB was more likely to occur than spontaneous PTB. Moreover,

Table I. Demographic, pregnancy and birth characteristics 34–36 weeks $N = 2637$.

	SPTL	PPROM	Indicated PTB	
	$N = 833$	$N = 641$	N = 1163	*p* value
Age ≥35 years	115 (13.8)	99 (15.4)	251 (21.6)	<0.001
Maternal ethnicity				<0.001
Caucasian	513 (61.6)	442 (69.0)	817 (70.2)	
African	7 (0.8)	8 (1.2)	21 (1.8)	
Asian	85 (10.2)	56 (8.7)	86 (7.4)	
Aboriginal	158 (19.0)	93 (14.5)	156 (13.4)	
Other	70 (8.4)	42 (6.6)	83 (7.1)	
History of PTB	27 (3.2)	21 (3.3)	20 (1.7)	0.047
Nulliparous	325 (39.0)	323 (50.4)	462 (39.7)	<0.001
Maternal smoking	238 (28.6)	160 (25.0)	292 (25.1)	0.161
Drug use	13 (1.6)	15 (2.3)	35 (3.0)	0.112
ART[a]	23 (2.8)	12 (1.9)	108 (9.3)	<0.001
Pre-eclampsia[b]	27 (3.2)	13 (2.0)	324 (27.9)	<0.001
Gestational diabetes	65 (7.8)	37 (5.8)	124 (10.7)	0.001
Placental abruption	10 (1.2)	4 (0.6)	24 (2.1)	0.038
Other APH[c]	95 (11.4)	50 (7.8)	79 (6.8)	0.001
Cervical circlage	10 (1.2)	8 (1.2)	11 (0.9)	0.794
Labour onset				<0.001
Spontaneous	833 (100)	349 (54.4)	0 (0)	<0.001
Induced	0 (0)	226 (35.3)	633 (54.4)	
No labour	0 (0)	66 (10.3)	530 (45.6)	
Mode of delivery				<0.001
Spontaneous vaginal	573 (68.8)	419 (65.4)	397 (34.1)	
Assisted vaginal	92 (11.0)	73 (11.4)	76 (6.5)	
Caesarean Section	168 (20.2)	149 (23.2)	690 (59.3)	
Birthweight (g)[d]	2650 (2619–2681)	2566 (2533–2599)	2570 (2534–2607)	0.001

[a]Assisted reproductive technology; [b]the onset of hypertension with proteinuria after the 20th week of gestation; [c]Any antepartum haemorrhage excluding placental abruption or placenta praevia; [d]Mean (95% CI).

Table II. Demographic, pregnancy and birth characteristics 27–33 weeks, $N = 1610$.

	SPTL	PPROM	Indicated PTB	
	$N = 478$	$N = 512$	$N = 620$	*p* value
Maternal age ≥35 years	86 (18.0)	105 (20.5)	146 (23.5)	0.077
Maternal ethnicity				0.069
Caucasian	341 (71.3)	374 (73.0)	439 (70.8)	
African	2 (0.4)	5 (1.0)	13 (2.1)	
Asian	41 (8.6)	25 (4.9)	47 (7.6)	
Aboriginal	70 (14.6)	84 (16.4)	84 (13.5)	
Other	24 (5.0)	24 (4.7)	37 (6.0)	
History of PTB	15 (3.1)	21 (4.1)	15 (2.4)	0.274
Nulliparous	231 (48.3)	227 (44.3)	313 (50.5)	0.117
Maternal smoking	119 (24.9)	150 (29.3)	125 (20.2)	0.002
Drug use	24 (5.0)	26 (5.1)	35 (5.6)	0.873
ART[a]	11 (2.3)	8 (1.6)	31 (5.0)	0.002
Pre-eclampsia[b]	8 (1.7)	11 (2.1)	330 (53.2)	<0.001
Gestational diabetes	23 (4.8)	29 (5.7)	31 (5.0)	0.812
Placental abruption	28 (5.9)	10 (2.0)	50 (8.1)	<0.001
Other APH[c]	118 (24.7)	87 (17.0)	50 (8.1)	<0.001
Cervical circlage	14 (2.9)	21 (4.1)	10 (1.6)	0.040
Labour onset				<0.001
spontaneous	478 (100)	347 (67.8)	0(0)	
induced	0 (0)	67 (13.1)	108 (17.4)	
No labour	0 (0)	98 (19.1)	512 (82.6)	
Mode of delivery				<0.001
Spontaneous vaginal	328 (68.6)	291 (56.8)	50 (8.1)	
Assisted vaginal	34 (7.1)	27 (5.3)	9 (1.5)	
Caesarean Section	116 (24.3)	194 (37.9)	561 (90.5)	
Birthweight (g)[d]	1740 (1698–1784)	1754 (1716–1791)	1515 (1474–1555)	<0.001

[a]Assisted reproductive technology; [b]the onset of hypertension with proteinuria after the 20th week of gestation; [c]Any antepartum haemorrhage excluding placental abruption or placenta praevia; [d]Mean (95% CI).

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women who experienced a medically indicated PTB at a late gestational age were more likely to be aged over 35 and to be of Caucasian ethnicity, whereas demographic characteristics were similar between the three clinical phenotypes at earlier gestational ages. ART was significantly more likely to be utilized in pregnancies resulting in indicated PTB at gestational ages greater than 27 weeks than those with a spontaneous PTB or PPROM. Pre-eclampsia was the principal pregnancy complication experienced by indicated PTBs at all gestational ages, consistent with previous studies [4].

The distribution of PTB phenotype has not been reviewed recently as far as we are aware. Further, comparisons with previous data are difficult due to differences in PTB definition [14,15]. Our findings are substantially different from previous estimates with 10% greater births for indicated reasons and fewer with idiopathic onset. This finding may, in part, result from the referral nature of the hospital. On the other hand, it may also reflect the increase, in developed countries, of late PTBs and suggests a systemic lowering of the threshold for early intervention for complications of pregnancy. Recent findings of significant morbidity among late PTBs compared with term births suggest that this change in practice may have deleterious impact on the short- and long-term costs of PTB [5,7].

Table III. Demographic, pregnancy and birth characteristics 23–26 weeks ($N = 275$).

	SPTL	PPROM	Indicated PTB	
	$N = 123$	$N = 88$	$N = 64$	p value
Maternal age ≥35 years	19 (15.4)	25 (28.4)	17 (26.6)	0.052
Maternal ethnicity				0.031
Caucasian	90 (73.2)	67 (76.1)	44 (68.8)	
African	2 (1.6)	1 (1.1)	4 (6.3)	
Asian	9 (7.3)	4 (4.5)	5 (7.8)	
Aboriginal	10 (8.1)	15 (17.0)	9 (14.1)	
Other	12 (9.8)	1 (1.1)	2 (3.1)	
History of PTB	2 (1.6)	2 (2.3)	2 (3.1)	0.799
Nulliparous	67 (54.5)	41 (46.6)	27 (42.2)	0.239
Maternal smoking	28 (22.8)	20 (22.7)	14 (21.9)	0.989
Drug use	7 (5.7)	5 (5.7)	3 (4.7)	0.954
ART[a]	1 (0.8)	1 (1.1)	0 (0.0)	0.710
Pre-eclampsia[b]	1 (0.8)	2 (2.3)	30 (46.9)	<0.001
Gestational diabetes	2 (1.6)	1 (1.1)	1 (1.6)	0.955
Placental abruption	12 (9.8)	6 (6.8)	13 (20.3)	0.027
Other APH[c]	38 (30.9)	21 (23.9)	6 (9.4)	0.005
Cervical circlage	9 (7.3)	5 (5.7)	3 (4.7)	0.757
Labour onset				<0.001
spontaneous	123 (100)	67 (76.1)	0 (0)	
induced	0 (0)	5 (16)	7 (10.9)	
No labour	0 (0)	16 (18.2)	57 (89.1)	
Mode of delivery				<0.001
Spontaneous vaginal	79 (64.2)	46 (52.3)	6 (9.4)	
Assisted vaginal	7 (5.7)	6 (6.8)	0 (0)	
Caesarean Section	37 (30.1)	36 (40.9)	58 (90.6)	
Birthweight (g)[d]	784 (750–817)	747 (714–781)	683 (637–728)	0.001

[a]Assisted reproductive technology; [b]the onset of hypertension with proteinuria after the 20th week of gestation; [c]Any antepartum haemorrhage excluding placental abruption or placenta praevia; [d]Mean (95% CI).

The finding in this study of a high prevalence of indicated PTBs at late gestational age should be viewed with caution in the context of the study setting. Midwife birth notification data demonstrated that only 40% of late PTBs in the state occurred at the study hospital. Late PTBs at a tertiary centre may not be representative of those at secondary or district hospitals. Nonetheless this should not influence the significant associations found in this study between indicated PTB and fertility treatment and maternal age. Similarly, the lack of comparison with term births could be considered a limitation of these data; this has been well characterised previously, however. It was our intention in this study to investigate characteristics of women with specific PTB phenotypes with an aim to target these groups for specialised care and attention.

The relationship between indicated PTB and PE was highly significant. While this finding appears to be self-evident, the strength of the data acts as a positive control confirming the integrity of the analysis. Investigation of the recognized heterogeneity within medically indicated PTB subtypes was beyond the scope of the present study; however this remains an important direction for future research.

An interesting finding was that a small proportion of women who had either SPTL or PPROM also had PE at all gestational ages, supporting emerging evidence that vascular and inflammatory changes associated with PE may also have a role to play in the pathogenesis of preterm labour [21,22]. Although the numbers are small, very preterm infants delivered for severe maternal illness such as PE consume significant resources in the intensive care nursery.

Assisted reproductive treatment was strongly associated with indicated PTB in this study. There is ample evidence that infertility has a strong relationship with PTB in general [16,23]. The strong association, demonstrated in the present study, between assisted reproductive treatment (ART) and indicated PTB in contrast to idiopathic PTB does not appear to have been described in other settings, however. These findings suggest that prevention of conditions leading to medically indicated PTB, in particular, should target women undergoing ART.

Bleeding during pregnancy was strongly associated with spontaneous PTB in this study. This finding was consistent with a recent study which found associations between PTB and multiple indicators of haemorrhage [24]. Previous studies have consistently shown that the risk of PTB is greater in women with a history of previous PTB [17]. In contrast, independent associations with previous obstetric history were not found with any of the three PTB phenotypes in the present study. The present data suggest that, while history of PTB influences the risk of PTB in general, this occurs consistently between the individual PTB phenotypes.

Maternal tobacco smoking is a recognized risk factor for PTB in many studies; however the mechanism of this effect remains unclear [8,13]. The present study found a significant association of smoking in only one PTB subtype: PPROM between 27 and 33 weeks. These data indicate that tobacco smoking has a specific effect on the fetal membranes while not influencing spontaneous labour or indicated PTB and suggest that interventions should target this outcome in future. Furthermore, smoking appears to have an impact on PPROM in a small gestational age window suggesting that research directed at this gestational age may be more likely to clearly elucidate the specific mechanisms of smoking on PTB.

There is ample evidence from the USA that African–American women are at higher risk of preterm birth than Caucasians, with

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or without accounting for socio-economic disparities [8,25]. Similarly, we found that non-Caucasian women in Australia are over-represented among spontaneous PTBs. Recent increases in PTB prevalence cannot be explained by changes in ethnic distributions, however; since the majority of PTBs in our region now occur for medically indicated reasons, primarily in Caucasian women.

Consistent with recent epidemiological research [1,3], the present study found that rapid increases in PTB in developed countries may be explained by increases in indicated PTB at gestational ages between 34 and 36 weeks. These data suggest a reducing threshold for tolerance of maternal and fetal complications in recent years. Emerging evidence that late PTB may have a significant adverse effect on neonatal health and long-term sequelae suggests that a return to watchful management of pregnancy complications rather than early delivery at this late gestational age may partially ameliorate these problems.

The distinct associations with specific PTB phenotypes described in this study offer the potential for targeted programs to prevent PTB. For example, women aged over 35 with a history of infertility are over-represented in medically-indicated PTB at later gestational ages. This information is valuable for counseling these women about expected outcomes for their pregnancies. Further, this subset of women would be the ideal target population to evaluate potential interventions to decrease the prevalence of late PTB.

Conclusions

Extreme PTB has the greatest individual impact on families and the healthcare system. In the vast majority of these, the aetiology of the PTB is idiopathic. In contrast, the greatest numbers of PTBs occur at late gestational age for medical indications. Consequently, while each individual late PTB costs less than an early PTB, overall these large numbers represent a significant cost burden of PTB to the medical system. These women are more likely to be older, Caucasian and requiring fertility assistance. At the present time the most viable strategy to lower the rates of PTB may be to target interventions to this high-risk population.

Carefully defined phenotypes and standardized definitions are the keys to effective assessment of PTB. These data highlight the importance of targeting our interventions if we are to have any success in reducing the rate of PTB in future.

***Declaration of interest*:** The authors report no declarations of interest.

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